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# TWO-ZONE MODEL OF LASER-INDUCED PLASMA

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*An algorithm modeling a plasma in two-zone source approximation and its interface with the NLopt library for multiparameter optimization were established. In order to allow the spectrum modeling algorithm and the NLopt library to work together an abstraction layer that initializes both libraries in one action, calculates a given type of loss function, and transfers its value to the optimization algorithm is used. The joint use of these algorithms was tested on model data; convergence to the plasma parameters used to obtain the synthetic test spectrum was achieved. It was shown that the two-zone model can be used to describe both ionic and atomic lines correctly, including those that undergo self-reversal during evaporation of the aluminum alloys, while methods of "blind" optimization of the loss function can be used to estimate the temperature and electron density in the laser-induced plasma from its spectra.*

***Keywords****: laser spark emission spectroscopy, plasma modeling, stochastic optimization.*

**Introduction.** Laser spark emission spectrometry (LSES) is an actively developing express method of direct atomic emission analysis based on recording the radiation of a laser-induced plasma formed by the action of powerful pulsed laser radiation focused on a sample. The unique potentials of LSES are used for preliminary assessment of the composition of the surface of Mars *in situ* (Mars rovers Curiosity and Perseverance [1, 2]) and the ocean floor at depths of up to 3000 m [3] and for various in-line industrial systems, including the analysis of molten metal or slag [4, 5]. Since the production of standard samples is impossible in all of the listed applications (the analysis of cosmic bodies, exploration of the seabed, control of various products and high-temperature melts) it is necessary to carry out research aimed at creating methods for determining the composition without reference samples using LSES. Existing analytical options without the construction of a calibration curve (calibration-free LIBS) require difficult conditions — the absence of self-absorption in the laser plasma [6] or total diagnostics (tomography) of the laser plume using the Abel [7] or Radon [8] transform with high spatial resolution. This greatly complicates the design of the spectral instrumentation, and a large number of plasma measurements is required. In terms of this approach to determination of the composition the best results were achieved by using a hydrodynamic model of an expanding laser plasma. However, this model does not have advantages in practice [9] in that it is too complex on account of the large number of parameters involved in the system of differential equations. These versions of standard-free analysis have low accuracy: the errors often reach 50% even for the major components [10]. Another approach to analysis without the use of reference samples is multiparameter optimization for approximation of the experimental spectra with model spectra based on a static model of a homogeneous laser plasma in a state of local thermodynamic equilibrium [11–15]. In this case, both the parameters of the plasma (temperature, electron density, size) and its elemental composition act as variable quantities. A significant problem with this model is the fundamental impossibility of modeling self-inversion of the strongest lines of the base components, which can lead to a solution with incorrect parameters and, consequently, to significant systematic error.

The aim of the present work was to check the adequacy of representation of the temperature and electron density gradients in laser-induced plasma in the form of two homogeneous zones, each of which has specific dimensions, and to assess the possibility of using the model to approximate the experimental spectra of the laser plasma.

**Methods**. In order to realize the possibility of modeling the spectra of a two-zone radiation source the previously described algorithm [13] was modified. Like the original version of the algorithm it was realized in a dynamic library

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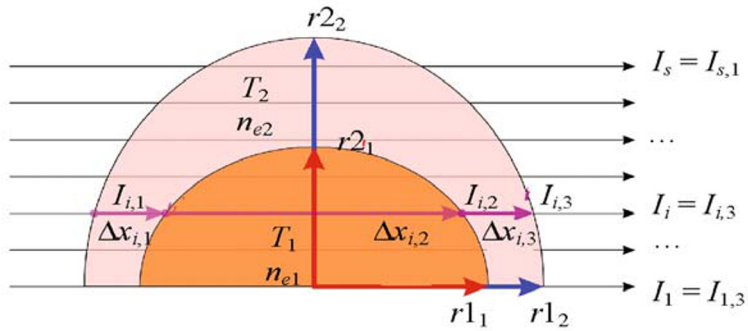


Fig. 1. Model of a two-zone laser-induced plasma viewed from the side: the zone boundaries are semi-ellipses with radii $r1$, $r2$ and with plasma parameters in each zone ($T$, $n_e$).

developed in C++. The library exports the definitions of classes whose functionality is hidden from the user, functions for the auxiliary operations (for example, retrieving necessary data on the transitions of the atoms and ions from the database) and calculations, as well as data structures in which the parameters of the plasma, apparatus, and calculation results are stored. The functional here is modified in such a way that the number of zones $z$ ($z \geq 1$) describing the source, the radii of the semi-ellipses $r1$ and $r2$ that determine the boundaries of the zones, and the values of temperatures $T$ and electron densities $n_e$ in them are transferred to the algorithm (Fig. 1). In height (along the axis of the laser beam) the plasma is divided into a given number $s$ of horizontal sections (layers) with identical thickness $h = r2_z/s$. The elemental composition (the array of the mass fractions of the elements $\boldsymbol{\omega} = \{\omega_k\}$) is assumed to be the same at all points of the plasma. Inside each $i$th layer ($i$ = 1, …, $s$) along the observation axis the radiation transfer equation [16] for a one-dimensional source is integrated recurrently:

$$\begin{cases} I_{i,1} = B_\lambda\left(T_z\right)\left(1 - e^{-\kappa_z(\lambda)\Delta x_{i,1}}\right), \\ I_{i,l} = I_{i,l-1}e^{-\kappa_{|z-l|+1}(\lambda)\Delta x_{i,l}} + B_\lambda\left(T_{|z-l|+1}\right)\left(1 - e^{-\kappa_{|z-l|+1}(\lambda)\Delta x_{i,l}}\right), \; l \geq 2, \; l \in N, \end{cases} \tag{1}$$

where $l$ is the number of the section of the optical path enclosed between the boundaries of the zones or the zone ($l$ = 1, …, $2z_i - 1$, $z_i$ is the number of zones that intersect the $i$th layer); $\Delta x_{i,l}$ is the length of this section (m); $\kappa_{|z-l|+1}(\lambda)$ is the absorption coefficient dependent on the wavelength $\lambda$ (m$^{-1}$); $B_\lambda(T_{|z-l|+1}])$ is the intensity of black body radiation [W/(sr·m$^2$·m)] given by Planck′s law [16], and $I_{i,l}$ is the intensity of radiation after passing the $l$th segment [W/(sr·m$^2$·m)]. The result is averaged over all the layers in order to obtain the final value of the intensity: $I = \Sigma I_i/s$. In this way we model the plasma with one, two, or more zones with the determined parameters.

To simplify the model to the one-dimensional case ($s$ = 1) the transfer of the radiation is only calculated along the base of the ellipses, the resulting intensity is $I = I_1$ (Fig. 1), and the number of model parameters is reduced by the number of zones. For the multi-zone plasma model of given composition we thus have 4 × $z$ parameters in the general case or 3 × $z$ parameters in the simplified case.

The absorption coefficients $\kappa(\lambda)$ in a specific zone depend on the volume concentrations of the atoms and ions, which can be calculated by simultaneous solution of the Saha equation [17] for each element and the equation of electroneutrality of the plasma [13]:

$$Vn_e = \sum_{k=1}^{K}\sum_{j=1}^{J_k} j\,\beta_{kj}(n_e, T)N_k = \sum_{k=1}^{K}\sum_{j=1}^{J_k} j\,\beta_{kj}(n_e, T)\,\frac{m\omega_k}{M_k}\,N_A = mN_A\sum_{k=1}^{K}\frac{\omega_k}{M_k}\sum_{j=1}^{J_k} j\,\beta_{kj}(n_e, T)\,, \tag{2}$$

where $V$ is the volume (cm$^3$); $n_e$ is the electron density (cm$^{-3}$); $m$ is the mass of the plasma zone (g); $N_A$ is the Avogadro number (mole$^{-1}$); $N_k$ is the number of particles of element $k$ in the plasma zone; $M_k$ is the molecular mass of element $k$ (g/mole); $j$ is the charge of the ion, and $\beta_{kj}(n_e, T)$ is the relative content of the ion of element $k$ with charge $j$ in the plasma zone, calculated from the system of Saha equations. By expressing the volume of the plasma zone as the ratio of the right side of Eq. (2) and $n_e$ it is possible to calculate the volume concentration of the particles ($n_k$, cm$^{-3}$) of element $k$:

$$n_k = \frac{N_k}{V} = \frac{n_e \omega_k}{M_k \sum_{k=1}^{K} \frac{\omega_k}{M_k} \sum_{j=1}^{J_k} j\beta_{kj}\left(n_e, T\right)} . \tag{3}$$

The number of particles in unit volume is required for calculation of the absorption coefficient of the individual spectral lines and the overall absorption spectrum (with correction for the enforced radiation):

$$\kappa\left(\lambda, n_e, T, \omega\right) = \sum_{k,j,m} \frac{\lambda_m^4}{8\pi c} A_{mn}^{(k,j)} \beta_{kj}\left(n_e, T\right) 10^6 n_k \frac{g_m^{(k,j)}}{Z^{(k,j)}\left(T\right)} \left(e^{-E_n^{(k,j)}/T} - e^{-E_m^{(k,j)}/T}\right) P_m^{(k,j)}\left(\lambda, n_e, T\right), \tag{4}$$

where $\lambda_m$ is the central wavelength of the transition (m) (the subscripts $k$ and $j$ for $\lambda_m$ are omitted); $c$ is the speed of light in the medium (m/s); $A_{mn}$ is the probability of the transition ($s^{-1}$); $g_m$ is the statistical weight of the upper level; $Z^{(kj)}$ is the sum among the states of element $k$ in the ionic state $j$; $E_m$ and $E_n$ are the energies of the highest and lowest levels (eV); $T$ is the temperature of the plasma (eV), and $P_m(\lambda)$ is the Voigt profile of the spectral line ($m^{-1}$), which takes account of Doppler and Stark broadening.

The bulk densities of the particles and the absorption spectra in each zone were calculated by means of Eqs. (3) and (4). If it is assumed that the plasma has axial symmetry in relation to the axis of the laser beam it is possible to calculate the volume of the central zone (subscript 1):

$$V_1 = \frac{2}{3} \pi r1_1^2 r2_1 \tag{5}$$

and of the remaining zones (subscript $p > 1$)

$$V_p = \frac{2}{3} \pi \left(r1_p^2 r2_p - r1_{p-1}^2 r2_{p-1}\right), \tag{6}$$

and also from (2) to find the mass of the atomized substance in each zone.

At the last stage of the algorithm the spectrum was convolved with the hardware function of the spectrograph and the electron-optical converter (EOC) and also the spectrum was averaged in line with the spectral width of the CCD matrix pixel (the final number of points in the model spectrum is equal to the width of the matrix in pixels).

During comparison of the experimental spectra with the model spectra linear interpolation of the experimental spectrum was used in order to remove the nonlinear calibration factor according to the wavelengths of any of the experimental data and also to subtract the background by the smallest amount and normalize the intensity to the maximum so that the dimensionless values of the experimental ($y$) and model ($\hat{y}$) spectra were in the range of [0; 1]. The regression problem was solved by means of the library of nonlinear multiparameter optimization algorithms NLopt [18], by minimizing the sum of squared errors (SSE):

$$\mathrm{SSE} = \Sigma_i (y_i - \hat{y}_i)^2 , \tag{7}$$

where $i$ is the number of the point in the spectrum.

In addition the average relative deviation between $y$ and $\hat{y}$ was calculated:

$$\mathrm{rSSE} = \sqrt{\Sigma_i (y_i - \hat{y}_i)^2 / \Sigma_i y_i^2} . \tag{8}$$

The library contains a large number of gradient and gradient-free optimization algorithms and also presents a unified interface to them. In order to allow the spectrum modeling algorithm and the NLopt library to work together an abstraction layer that initializes both libraries into one action, calculates a given type of loss function, and transfers its value to the optimization algorithm is used.

**Results and Discussion**. The most important task in multiparameter optimization is to minimize the calculation time, i.e., to find a compromise between the minimum loss function and the number of iterations required to achieve it in the search for optimal parameters of the plasma. In order to select the best algorithm all the global gradient-free optimization

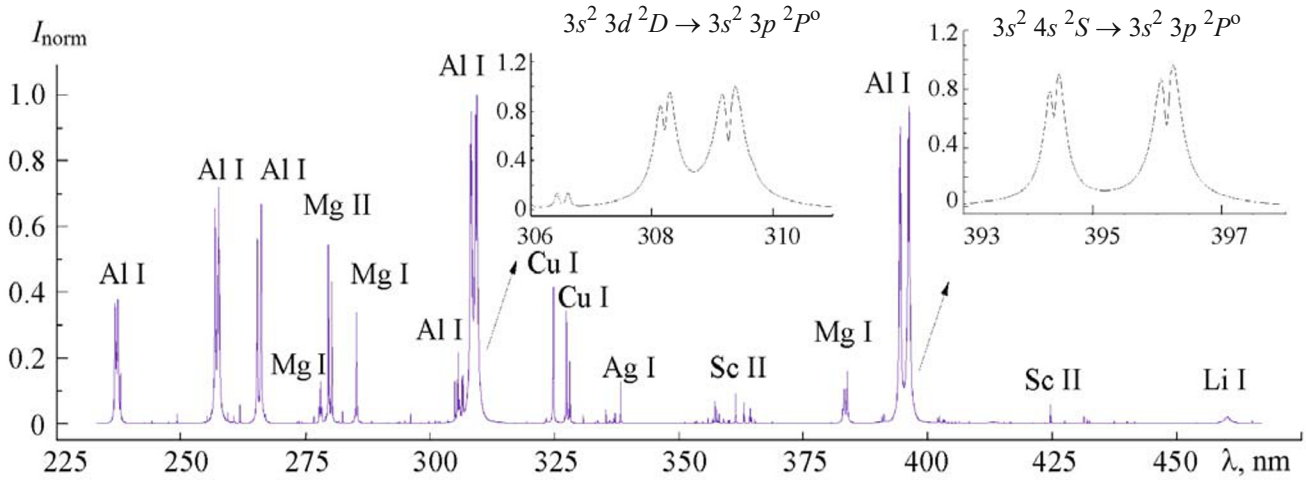


Fig. 2. Model (test) spectrum of aluminum alloy; the strongest lines are shown, the self-reversed lines of Al I are shown in the insets.

algorithms available in the NLopt library were applied to the experimental spectrum of the previously obtained stainless steel [13] over a relatively narrow spectral range of 392–412 nm. The simplest model was used to describe the spectrum (one plasma zone, one layer), the elemental composition of the plasma was set equal to the composition of the sample, and the parameters $T$, log $n_e$, and $r1 = r2$ were optimized. Only the DIRECT [19] and CRS2-LM [20, 21] algorithms achieved convergence of less than 1000 iterations with SSE ~ 0.6 (for 1390 points in the spectrum).

The DIRECT algorithm addresses a hypercube in parameter space and divides it sequentially into hyperparallelepipeds, at the center of each of which it calculates the value of the function. At each iteration from the set of available hyperparallelepipeds those that can potentially contain an optimum are selected (i.e., values lower than those of the other hyperparallelepipeds and/or having sufficiently large unexplored regions), and they are then divided further until the stopping criteria are met. In a series of successive runs the original version of DIRECT could go inexplicably into a loop and return values that have no physical meaning, and this algorithm had therefore to be abandoned.

The CRS2 algorithm [19] works with a population of points in parameter space initially selected at random ($10(N + 1)$ points are usually taken for the $N$ parameters). The point at which the value of the function to be minimized is the largest is selected at each iteration, and like the complex method this point is reflected in relation to the population centroid. In the modification with "local mutation" [20] if the reflection relative to the centroid did not lead to a decrease in the value of the function the point with the largest value of the function is tried in relation to the point with the smallest value of function with random weights at each coordinate. The point with the smaller of the two obtained values is included in the final population. The CRS2-LM algorithm does not have the inherent disadvantage of the DIRECT algorithm, and this made it possible to test it for approximation of the spectra within the scope of more complex plasma models.

In order to test the correctness of the solution to the inverse problem within the framework of the two-zone model ("hot" core and "cold" periphery of the plasma), i.e., the ability to achieve convergence for the parameters $T_1$, $T_2$, log $n_{e1}$, log $n_{e2}$, $r1_1$, $r1_2$–$r1_1$, $r2_1$, and $r2_2$–$r2_1$, the optimization algorithm was tested on the model spectrum of an aluminum alloy (wt.%): Cu 3.6, Li 1.06, Mg 0.59, Ag 0.4, Sc 0.1, Mn 0.06, obtained for the given parameters (Table 1) in the range 233–467 nm (Fig. 2), containing the distinct self-absorbed and self-reversed resonance lines of aluminum (transitions from the excited $3s^2ns$ and $3s^2md$ states to the ground $3s^23p$ state [22], where $n \geq 4$, $m \geq 3$) and also the weaker lines of the other elements (~300 lines). The contours of the lines with known Stark broadening parameters were calculated individually, and a value of 0.006 nm at $n_e = 10^{17}$ cm$^{-3}$ was used for the remaining lines. Weak lines were excluded from the calculation ($10^{-5}$ in relation to the strongest line without taking account of self-absorption), and the threshold for calculating the wings of the line amounted to $10^{-7}$ in relation to its maximum. The full width at the half-height of the instrumental function of the spectrometer with the parameters in [13] was ~0.065 nm. The plasma along the vertical coordinate was divided into $s = 40$ layers (Fig. 1).

After 30,000 iterations an almost ideal match was obtained between the spectra (SSE = $1\cdot10^{-5}$, rSSE = $3\cdot10^{-4}$) in addition to convergence of the parameters of the inner zone $T_1$, log $n_{e1}$, $r1_1$ and outer zone $T_2$ and log $n_{e2}$ within the specified

TABLE 1. Test Parameters for the Model Spectrum and the Results of Its Approximation

| Parameter | Given value (40 layers) | Lower search limit | Upper search limit | Convergence criterion | Optimized value (40 layers) | Relative deviation, % | Optimized value (one layer) |
|---|---|---|---|---|---|---|---|
| $T_1$, K | 8120 | 4640 | 23,210 | 10 | 8120 | <0.1 | 8120 |
| $T_2$, K | 3480 | 1160 | 11,600 | 10 | 3550 | 2.0 | 3450 |
| log ($n_{e1}$, cm$^{-3}$) | 16.900 | 15.000 | 18.000 | 0.001 | 16.899 | <0.006 | 16.897 |
| log ($n_{e2}$, cm$^{-3}$) | 14.300 | 13.000 | 18.000 | 0.001 | 14.298 | −0.014 | 14.109 |
| $r1_1$, mm | 1.00 | 0.01 | 3.00 | 0.01 | 1.005 | <1 | 0.81 |
| $r1_2$−$r1_1$, mm | 1.00 | 0.01 | 3.00 | 0.01 | 1.61 | 61 | 2.29 |
| $r2_1$, mm | 1.00 | 0.01 | 3.00 | 0.01 | 0.54 | −46 | – |
| $r2_2$−$r2_1$, mm | 2.00 | 0.01 | 3.00 | 0.01 | 0.64 | −68 | – |

limits. The average time of the calculation for one spectrum was 0.56 s (on one AMD Ryzen 2600X 3.8 GHz processor core). However, the plasma sizes $r1_2$–$r1_1$, $r2_1$, and $r2_2$–$r2_1$ did not converge to the specified values, which indicates ambiguity in the decision to use the layer-averaged and normalized spectrum. A model with one layer was therefore also tested, and in this case good agreement was achieved with the normalized intensity values (SSE = $4\cdot10^{-3}$, rSSE = $5\cdot10^{-3}$). All the parameters reached the established convergence limits after 5815 iterations, and the average time for calculating one spectrum was reduced to 0.21 s. The temperatures as well as the electron density in the inner zone converged to the given values (Table 1, the last column), but in the outer zone $n_e$ was underestimated by 35%. The dimensions along the base of the plasma (Fig. 1, the bottom layer) also diverged especially in the thickness of the outer zone (by more than130%). This probably results from compensation for layers that are absent in the calculations only for the outer zone (the ($r2_1$; $r2_2$) section along the plasma height). We note that the loss function reaches SSE = $4\cdot10^{-3}$ after ~2700 iterations in 40 layers while the $T_1$, log $n_{e1}$, $T_2$, log $n_{e2}$, and $r1_1$ values do not differ from the specified *a priori* values by more than 3%. The algorithm continues to vary the $r1_2$–$r1_1$, $r2_1$, and $r2_2$–$r2_1$ values and cannot meet the compatibility criteria in relation to them. In the case of a 2D model with several layers the obtained plasma size data should therefore be used with caution since the solution with respect to them is unstable.

The CRS2-LM approximation algorithm was tested in order to approximate the experimental spectrum of an aluminum alloy (390–400 nm) produced in air with a delay of 2 μs after the laser pulse, where the self-reversing resonance lines of Al I are present (Fig. 3), within the framework of one-band and two-band models with five layers. As expected, the single-band model does not make it possible to realize approximation of the contour of the self-reversing lines (rSSE = 0.14), which leads to increase of the temperature by 1200 K and of the electron density by 1.3 times in relation to the "hot" zone in the two-zone model (Table 2). As a result of this the contour of the low-intensity ion line of Al II 390.07 nm is described incorrectly by the one-zone model (Fig. 3).

The two-zone model, on the contrary, describes the self-reversing contour of the aluminum lines well in addition to the profiles of the other low-intensity lines (rSSE = 0.06). The plasma parameters at which compatibility of the model with the experimental spectrum is achieved (Table 2) also agrees well with existing ideas about the laser-induced plume on the surface of the first target: the small hot region at the center is surrounded by the cooling cloud of the expanding plasma, in which the temperature and electron density are significantly lower. We estimated the electron density of the plasma from the H-α 656.3 nm line by means of the equation $w_S = 4.63\cdot10^{-12} n_e^{2/3}$ [23] and obtained log $n_e$ = 17.15, which is close to the optimized value of 17.28 for the inner zone of the two-zone model (Table 2). As expected, the time for calculating one spectrum in the narrow spectral interval is an order of magnitude lower (0.02 s) than in the synthetic problem, but the number of iterations can still be large (up to 25,000) due to the ambiguity of the solution for the plasma sizes and particularly for $r2$. Random approximation errors were also estimated on the basis of 50 parallel runs with intensity randomization at each point of the experimental spectrum within the measurement error (for the spectrum in Fig. 3 RSD ~ 3%). As seen from Table 2, good stability in the convergence of the parameters in the two-zone model is only achieved for the $T_1$ and log $n_{e1}$ parameters, is satisfactory for $T_2$, and is as expected poor for $r2_1$ and $r2_2$.

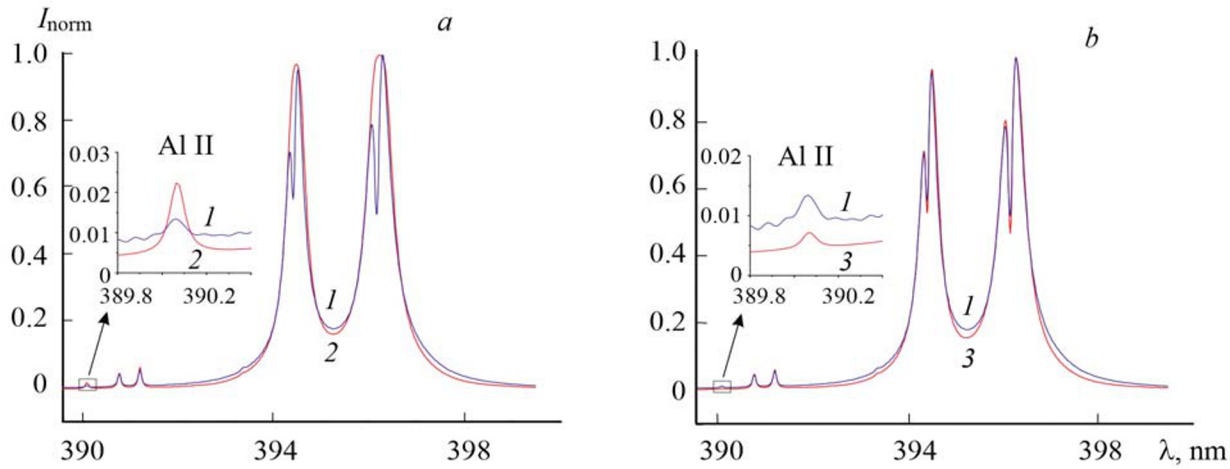


Fig. 3. The spectrum of the aluminum alloy (recording delay 2 μs) with self-reversed lines Al I 394.40 and 396.15 nm and its approximation by the one-zone (a) and two-zone models: 1) experiment; 2) model (one zone); 3) model (two zones); on the insets the weak line is Al II 390.07 nm.

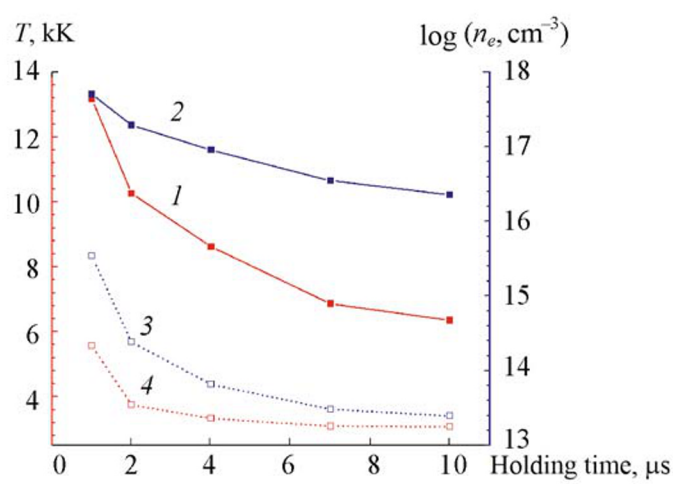


Fig. 4. Variation of the optical parameters during approximation of the spectra of aluminum alloy obtained with various holding times: $T_1$ (1), log $n_{e1}$ (2), log $n_{e2}$ (3), and $T_2$ (4).

TABLE 2. Results from Approximation of the Experimental Spectrum of the Aluminum Alloy (Fig. 3)

| Zone | $T_1$, K | $T_2$, K | log $n_{e1}$ | log $n_{e2}$ | $r1_1$, mm | $r1_2$, mm | $r2_1$, mm | $r2_2$, mm |
|---|---|---|---|---|---|---|---|---|
| 1 | 11,510 ± 60 | – | 17.397 ± 0.008 | – | 1.46 ± 0.07 | – | 6 ± 4 | – |
| 2 | 10,300 ± 700 | 3800 ± 800 | 17.28 ± 0.04 | 14.4 ± 0.9 | 1.4 ± 0.7 | 3.9 ± 0.7 | 1.2 ± 0.9 | 2.7 ± 1.6 |

In addition the adequacy of description of the laser plasma by means of the two-zone model was checked by approximating the experimental spectra obtained with different delay times. In this case the change in the optimal parameters ($T$, $n_e$), obtained by approximation with the two-zone model, corresponds to the physical process of expansion and cooling of the laser plume (Fig. 4). The two-zone model is thus a good approximation that makes it possible to describe the temperature

gradient and electron density in the plasma (to obtain an assessment of their values at the center and at the periphery) and to approximate the experimental spectra, including the self-reversing lines, with high accuracy.

**Conclusions.** The model spectra obtained with the two-zone model show that solution of the problem of multiparameter optimization of the laser-induced plasma parameters is compatible with the given values. The use of the given algorithms for determination of the properties is justified. Here it is possible mostly to achieve compatibility with use of the CRS2-LM algorithm from the "random search" set of methods. With use of the two-zone model it is possible to describe both ionic and atomic lines correctly, including those that are subject to self-reversal. The methods of "blind" optimization of the function of losses can be used to determine the temperature and electron density in laser-induced plasma from the spectra.

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